



SHADOWMAP

**AI-POWERED NETWORK RECONNAISSANCE & VULNERABILITY
INTELLIGENCE PLATFORM**

Prithak Das

INTRODUCTION

ShadowMap is a GUI-based cybersecurity reconnaissance tool built on top of Nmap, designed to perform comprehensive attack surface mapping and exposure analysis for any given target.

It extends traditional port scanning by integrating real-time vulnerability intelligence, OSINT enrichment, and AI-driven security insights into a unified, modern web interface.

Additionally,

- To enhance decision-making, ShadowMap leverages LLaMA 3 via Groq to generate actionable remediation recommendations instantly after scan completion.
- The platform integrates multiple OSINT sources, allowing users to directly explore external intelligence related to the target, making reconnaissance faster, more contextual, and more efficient.



FEATURES

OSINT & RECONNAISSANCE

- **OSINT Links:** Direct access to platforms like VirusTotal, IPVoid, Shodan and more
- **Subdomain Discovery:** Real-time enumeration using DNSDumpster

CORE FEATURES

- **AI Insights:** Post-scan analysis powered by LLaMA 3 (Groq) delivering actionable security recommendations.
- **Intent Router:** Smart query handling with predefined responses, CVE parsing, and AI fallback.
- **ShadowAI Chat:** Persistent in-app assistant with context awareness and localStorage support.

VULNERABILITY INTELLIGENCE

- **CVE Enrichment:** Real-time vulnerability mapping via CPE lookup from National Vulnerability Database (minimizes false positives)
- **Danger Score:** Dynamic risk scoring model: $\text{Critical} \times 35 + \text{High} \times 20 + \text{Medium} \times 10$
- **Shodan Integration:** Live host intelligence using Shodan (ISP, org, domain name/IP, banners)

SCANNING & ANALYSIS

- **Nmap Integration:** Supports Quick, Aggressive, and fully Custom scan profiles using Nmap
- **NSE Script Support:** Browse and execute installed Nmap NSE scripts dynamically
- **Custom Flags Engine:** 70+ Nmap flags categorized into 8 groups with searchable multi-select UI

REPORTING & DATA MANAGEMENT

- **Scan History:** Stores last 10 days of scans with per-entry deletion and detailed reports
- **Export Reports:** Download reports as PDF (print-ready) or XHTML

SYSTEM IN ACTION

 TARGET SELECTION

IPV4 / HOSTNAME

e.g. 192.168.1.1

PARAMETERS

 QUICK SCAN

 AGGRESSIVE SCAN

 STEALTH (CUSTOM)

CUSTOM PORTS

e.g. 80,443 or 1-1000

NMAP NSE SCRIPTS

Search script...

EXECUTE

User Input Interface

22	SSH	OpenSSH 8.0 protocol 2.0	CRITICAL CVSS 9.8	CVE-2023-38408 The PKCS#11 feature in ssh-agent in OpenSSH before 9.3p2 has an insufficiently t... 9.8 NVD CVE-2008-3844 Certain Red Hat Enterprise Linux (RHEL) 4 and 5 packages for OpenSSH, as signed ... 9.3 NVD CVE-2026-35385 In OpenSSH before 10.3, a file downloaded by scp may be installed setuid or setg... 8.1 NVD CVE-2026-35386 In OpenSSH before 10.3, command execution can occur via shell metacharacters in ... 8.1 NVD CVE-2026-35414 OpenSSH before 10.3 mishandles the authorized_keys principals option in uncommon... 8.1 NVD
25	SMTP	-	LOW	No known CVEs
53	DOMAIN	PowerDNS Authoritative Server 4.9.5	CRITICAL CVSS 9.8	CVE-2026-33608 An attacker can send a notify request that causes a new secondary domain to be a... 9.8 NVD CVE-2026-33257 An attacker can send a web request that causes unlimited memory allocation in th... 7.5 NVD CVE-2026-33260 An attacker can send a web request that causes unlimited memory allocation in th... 7.5 NVD

Vulnerability Analysis (CVE Detection)

ShadowMap Vulnerability Report

SCAN COMPLETE

DOWNLOAD

SCANNER

DARK

Target

45.118.162.239

Ports Found

12

Services

7

Scan Time

315.67s

Location

Mumbai, India (Yotta Network Services Private Limited)

Danger Score

LOW

100

Critical

HIGH

Critical

4

High

0

Medium

0

Low

8

Risk Scoring Dashboard

Target Information

Domain: brainwareuniversity.ac.in

Hostnames: brainwareuniversity.ac.in

Location: Mumbai, India (Yotta Network Services Private Limited)

ISP: Yotta Network Services Private Limited

Operating System: Linux 3.10 - 4.11

Scan Type: Aggressive

Status: Online

Target Intelligence Overview

Subdomains

admission.brainwareuniversity.ac.in

www.admission.brainwareuniversity.ac.in

alumni.brainwareuniversity.ac.in

www.alumni.brainwareuniversity.ac.in

brainwave-journal.brainwareuniversity.ac.in

Subdomain Enumeration Results

AI Insights

- ▶ The OpenSSH 8.0 protocol 2.0 service on Port 22 should be hardened by implementing key-based authentication, disabling password authentication, and limiting user access to reduce the risk of brute-force attacks.
- ▶ The PowerDNS Authoritative Server 4.9.5 on Port 53 should be configured to only allow queries from trusted sources, and DNSSEC should be enabled to prevent DNS spoofing and amplification attacks.
- ▶ The Apache httpd services on Ports 80 and 443 should be hardened by disabling unnecessary modules, configuring secure protocol versions (TLS 1.2 and 1.3), and implementing a web application firewall (WAF) to protect against common web attacks.

AI-Powered Security Insights

SYSTEM ARCHITECTURE



- OSINT Enrichment (Shodan)
- Vulnerability Lookup (NVD)
- AI Insights (Groq)

TECHNOLOGY STACK

BACKEND

Python 3.10+ – Core application logic and orchestration

Flask 3.x – Lightweight backend framework for API and routing

AI & INTELLIGENCE

LLaMA 3 (via Groq) – AI-powered vulnerability analysis, remediation suggestions, and interactive user assistance

VULNERABILITY DATA

National Vulnerability Database API – Real-time CVE enrichment using CPE-based matching

OSINT & EXTERNAL INTEGRATIONS

Shodan API – Host intelligence (ISP, org, open ports, banners)

DNSDumpster – Subdomain discovery

VirusTotal / SecurityTrails / IPVoid – External OSINT lookups

FRONTEND & UI

HTML, CSS, JavaScript – Core frontend stack

TomSelect 2.3 – Advanced searchable dropdown for Nmap flags and NSE scripts

SCANNING ENGINE

Nmap – Core engine for network scanning, service detection, and NSE scripting

REPORTING & DATA HANDLING

PDF (Print API) – Export scan reports

XHTML – Structured report generation

LocalStorage – Chat persistence and session data

SECURITY & CONTROL

Custom Flag Allowlist – Secure execution of Nmap commands

AbortController (JS) – Real-time scan cancellation

IMPACT



1. Centralized Reconnaissance Hub:

- Aggregates critical target intelligence—including IP details, subdomains, open ports, and CVEs—into a single unified platform for faster and more effective security analysis.

2. Simplifies Cybersecurity for Non-Experts:

- Provides an intuitive graphical interface for performing network scans without requiring command-line knowledge.
- Integrated AI assistance helps users understand scan types (e.g., Nmap flags like -sS), choose appropriate scanning methods, and interpret results effectively.

3. Automated CVE Mapping via NVD Integration:

- Leverages real-time integration with the National Vulnerability Database (NVD) to automatically map detected service versions to known CVEs, enabling instant identification and prioritization of vulnerabilities.

4. AI-Powered Vulnerability Insights:

- Utilizes AI to analyze detected CVEs and provide actionable recommendations for securing systems and reducing potential attack risks.

5. Real-Time Security Analysis:

- Delivers immediate insights into network exposure and vulnerabilities through automated scanning and analysis.

INNOVATION & UNIQUENESS



AI-DRIVEN DECISION SUPPORT

Beyond traditional scanning, ShadowMap leverages LLaMA 3 to analyze vulnerabilities, provide remediation suggestions, and explain security concepts, enabling both beginners and professionals to make faster, informed decisions.

PRECISION VULNERABILITY MAPPING

Utilizes CPE-based matching with the National Vulnerability Database (NVD) to accurately map services to real CVEs, significantly reducing false positives compared to keyword-based approaches.

ALL-IN-ONE RECON PLATFORM

Integrates Nmap scanning, OSINT intelligence, vulnerability analysis, and reporting into a single unified interface, eliminating the need for multiple tools.

LIMITATIONS

1. INCONSISTENT CHAT PERSISTENCE

- Conversations with ShadowAI are not synchronized between the index page and dashboard.
- Chat history is lost after page refresh or directory navigation in the dashboard, reducing continuity of user interaction.

2. CVE ACCURACY LIMITATIONS

- In certain cases, vulnerability results may include false positives or miss relevant CVEs due to inconsistencies or limitations in the NVD API response.

3. LIMITED DATA STORAGE

- The system currently stores only a limited number of scans and AI interactions for a short duration (~10 days).
- Lack of a robust database restricts long-term analysis and historical tracking.

4. DEPENDENCY ON EXTERNAL APIs

- The platform relies heavily on third-party APIs (NVD, Shodan, etc.).
- Any API failure, downtime, or rate limiting can directly impact scan results and system performance.

5. REVERSE LOOKUP INCONSISTENCIES

- Reverse DNS lookup may fail or return inaccurate domain mappings.
- This can affect subdomain discovery and reduce the completeness of reconnaissance results.

FUTURE ENHANCEMENT

PERSISTENT CHAT MANAGEMENT

- Implements centralized storage and session management to ensure seamless synchronization of ShadowAI conversations across all modules, maintaining chat continuity after refresh or navigation.
- **Implementation:** Uses cloud-based database with session IDs to store and retrieve chat history across pages.

IMPROVED CVE ACCURACY

- Enhances vulnerability detection using structured service-version mapping and validation to reduce false positives from NVD.
- **Implementation:** Applies version-based filtering with CPE mapping and cross-verification of CVE results before displaying.

SCALABLE DATA STORAGE

- Integrates a robust database system to store scan history, reports, and AI interactions for long-term access.
- **Implementation:** Uses scalable cloud databases with structured schemas to efficiently store and retrieve historical scan data.

RELIABLE API HANDLING

- Incorporates caching and fallback strategies to maintain consistent output during API failures or rate limits.
- **Implementation:** Implements local caching (Redis/local storage) and backup API sources to ensure uninterrupted data flow.

ENHANCED REVERSE LOOKUP MECHANISM

- Improves domain resolution accuracy using multiple DNS and OSINT sources.
- **Implementation:** Combines multiple lookup services (DNS, reverse IP tools) and validates results before final output.

**THANK
YOU**

